

Result 1: If $U_1, \dots, U_n \stackrel{\text{iid}}{\sim} U(0, 1)$, where $n = 2k+1$ is odd, the sample median $\tilde{U} \sim \text{Beta}(k, k)$

Result 2: $F_\mu^{-1}(\tilde{U})$ & $\tilde{X}$ have same distⁿ

$$f_x(x) = \begin{cases} \frac{1}{2} e^{x-\mu}, & x < \mu \\ 1 - \frac{1}{2} e^{-(x-\mu)}, & x \geq \mu \end{cases}$$

$$F_U^{-1}(u) = \begin{cases} \mu + \log 2u, & u < 1/2 \\ \mu - \log 2(1-u), & u \geq 1/2 \end{cases}$$

$$\tilde{X} = F_\mu^{-1}(\tilde{U}) = \begin{cases} \mu + \log 2\tilde{U}, & \tilde{U} < 1/2 \\ \mu - \log 2(1-\tilde{U}), & \tilde{U} \geq 1/2 \end{cases}$$

{result 2}

$$\tilde{X} - \mu = \begin{cases} \ln 2\tilde{U}, & 0 < \tilde{U} \leq 1/2 \\ -\ln 2(1-\tilde{U}), & 1/2 < \tilde{U} < 1 \end{cases}$$

$$\text{If } \tilde{X} - \mu < 0 \text{ then } F_{\tilde{X}-\mu}(\tilde{x}-\mu) = \mathbb{P}(\ln(2\tilde{U}) \leq \tilde{x}-\mu) = \mathbb{P}\left(\tilde{U} \leq \frac{e^{\tilde{x}-\mu}}{2}\right) \\ = F_{\tilde{U}}\left(\frac{e^{\tilde{x}-\mu}}{2}\right)$$

$$\text{If } \tilde{X} - \mu \geq 0 \text{ then } F_{\tilde{X}-\mu}(\tilde{x}-\mu) = \mathbb{P}(\ln(2\tilde{U}) \leq \tilde{x}-\mu, \tilde{U} \leq 1/2)$$

$$+ \mathbb{P}(-\ln 2(1-\tilde{U}) \leq y, \tilde{U} > 1/2)$$

$$= \mathbb{P}(\tilde{U} \leq 1/2) + \mathbb{P}(-\ln 2(1-\tilde{U}) \leq y, \tilde{U} > 1/2)$$

$$= \frac{1}{2} + \mathbb{P}\left(U \geq 1 - \frac{e^{-(\tilde{x}-\mu)}}{2}\right) - \frac{1}{2} = 1 - F_{\tilde{U}}\left(1 - \frac{e^{-(\tilde{x}-\mu)}}{2}\right)$$

$$C_n \geq 0,$$

$$\mathbb{P}(-C_n \leq \tilde{X} - \mu \leq C_n) = F_{\tilde{X} - \mu}(C_n) - F_{\tilde{X} - \mu}(-C_n) = 1 - \alpha$$

$$\Rightarrow 1 - F_{\tilde{U}}\left(1 - \frac{e^{-C_n}}{2}\right) - F_{\tilde{U}}\left(\frac{e^{-C_n}}{2}\right) = 1 - \alpha$$

$$F_{\tilde{U}}\left(\frac{e^{-C_n}}{2}\right) = \alpha/2 \quad \left\{ \because F_{\tilde{U}}(1-d) = F_{\tilde{U}}(d) \right\}$$

$$\frac{e^{-C_n}}{2} = F_{\tilde{U}}^{-1}(\alpha/2)$$

$$C_n = -\ln 2(F_{\tilde{U}}^{-1}(\alpha/2))$$

$F_{\tilde{U}}^{-1}(\alpha/2)$ is $(\alpha/2)^{th}$ quantile.